

# Multi-Agent Simulation and Crowdsourced Reviews for Urban Vitality Analysis: Insights from a Peri-Urban City

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# Agenda



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## Introduction

- Context
- Research Questions



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## Methodology and assumptions

- Methodological approach
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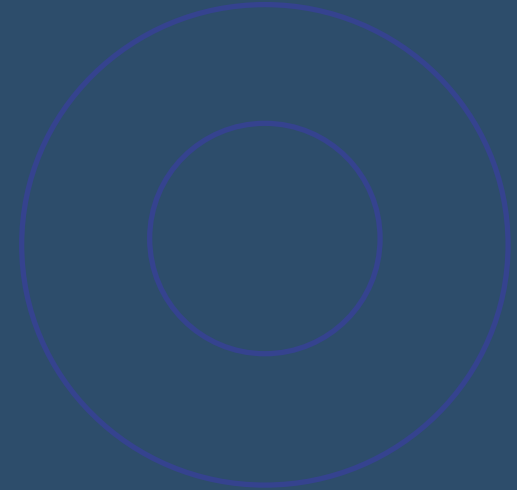
## Results

- Traffic flow
- Correlation analysis



04

## Conclusions and Perspectives



# Why focus on urban vitality?

Urban vitality refers to a lively, welcoming urban space that ensures growth, mobility, and diversity (Liu et al. 2022).



## Early Approach: Environmental psychology

Interviews and Surveys in face-to-face or online treating of satisfaction, environmental preference, quality of life, place attachment etc.(Lee et al. 2021)



## Access to new data: Area of Big data

Taxi data, mobile phone data, satellite imagery, points of interest (POIs), remote sensing, public transit maps, bike-sharing data + machine/deep learning (Amer et al. 2025)



## Trip flows shaping urban patterns

Multimodal flows (car, walking, cycling, transit) reveal the level of activity and occupancy in public spaces.

## Our approach

Use the MATSim multi-agent simulation framework to capture, at a fine granularity, trip dynamics and the interactions between users and urban planning.

## Study area

**Loos-en-Gohelle**, a former mining town in the Pas-de-Calais region, a typical example of peri-urban spatial fragmentation.

70%

Of the territory is agricultural

# A central question, A practical obstacle

**Question :** Are the flow patterns simulated by MATSim consistent with the actual appeal of public spaces, as revealed by digital traces (Google Maps reviews)?

The key challenge: The lack of counting data

In a peri-urban context like Loos-en-Gohelle, where traffic counts are unavailable, the authors use geolocated online review volumes as a proxy for actual traffic.

**6,799** inhabitants in Loos-en-Gohelle

**1.4 M** Inhabitants modeled  
in the Pas-de-Calais department

**6,706 km<sup>2</sup>** Modeled area in the  
Pas-de-Calais department

# Methodological approach

*Connecting synthetic populations with online reviews using MATSim simulations*



## 1. Generation of travel demand

The Eqasim framework (Hörl et al., 2021):  
Synthetic agents, activity chains, and spatial localization based on open sociodemographic data.



## 2. Dynamic simulation via MATSim

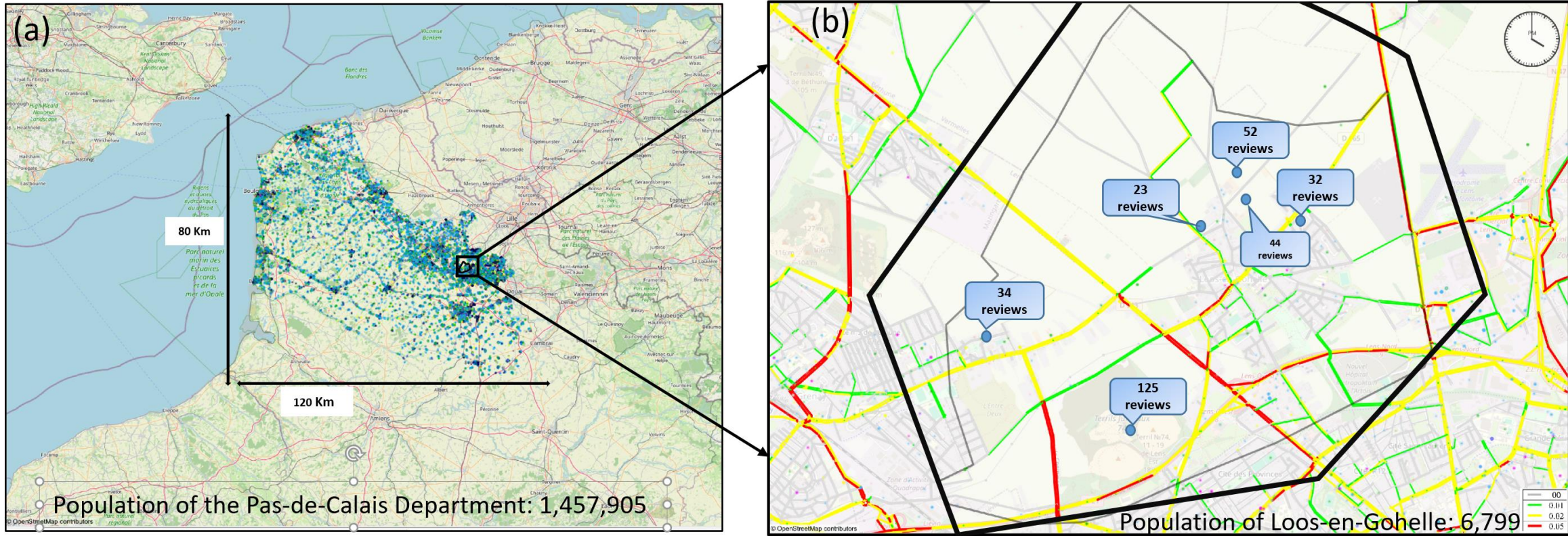
- Agent interactions on the network based on a calibrated scoring function.
- “Cutter” scenario for detail simulation in Loos-en-Gohelle.



## 3. Analysis and Correlation

- Data flows are extracted from MATSim event files and aggregated by cadastral section.
- Evaluate local attendance by correlating simulation data with online review volumes.

# From department scale to local scenario



**(a)** Department-level overview (1.4 million inhabitants) **(b)** Focus on Loos-en-Gohelle with selected online review volumes by POI

# Online Reviews as a Proxy for Urban Vitality

In the absence of traffic count data to validate the simulation, the study relies on geolocated Google Maps reviews as an alternative source.



## Automated Extraction

Data scraping was performed via a Python script using the Selenium package between March and April 2026.



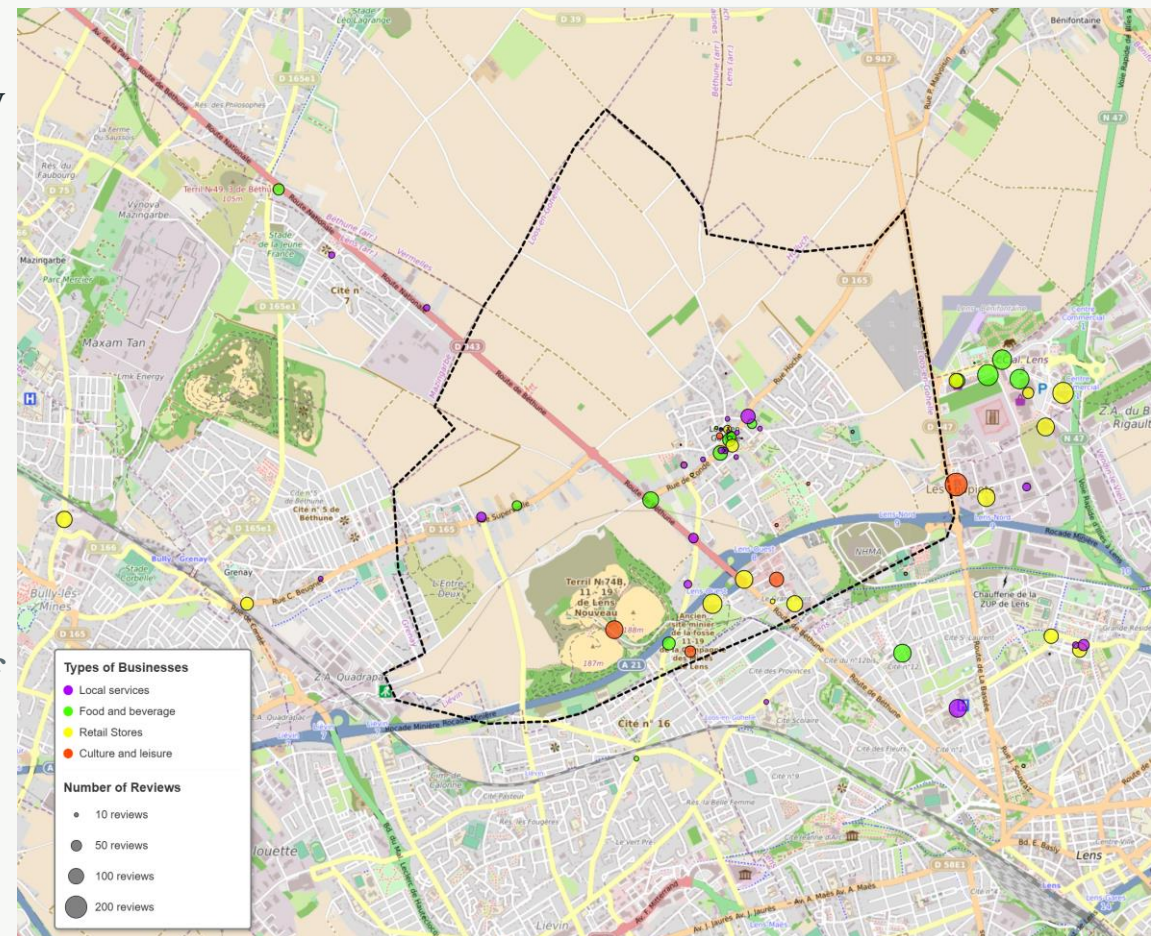
## Assumption (Li et al. 2025)

Each review reflects an actual in-person visit to a point of interest.



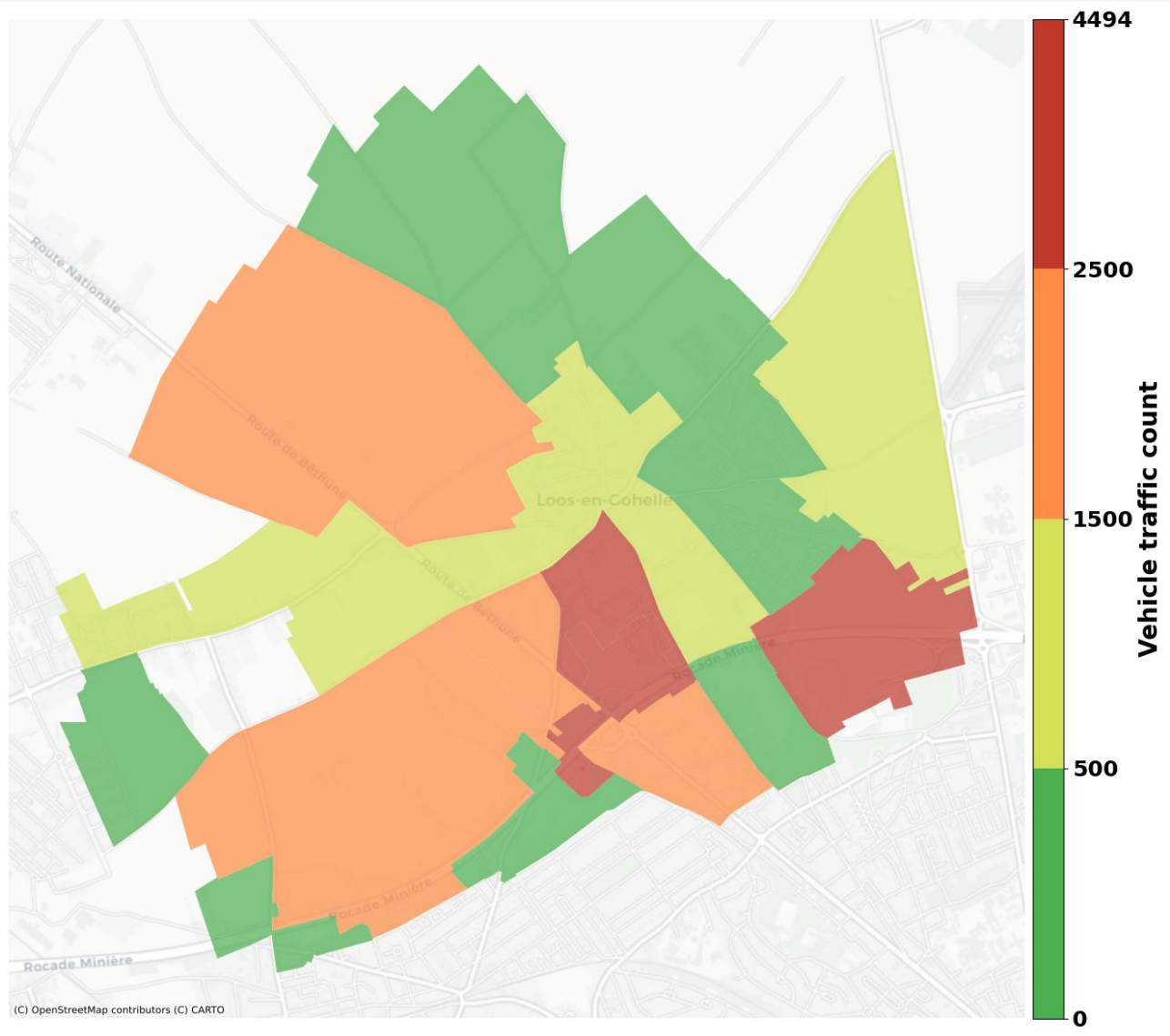
## Validated Proxy (Noufe et al. 2026)

Review volumes are a proven proxy for mapping urban vitality.



Online Review Volume by Business  
in Loos-en-Gohelle.

# Spatial Asymmetry in Traffic Flows



■ **Southeast: peaks in intensity**

> 2,500 vehicles along major thoroughfares (Route de Béthune, Rocade Minière), through and commercial traffic, excluding the historic center.

■ **West / Central-West residential areas**

High traffic volumes (1,500–2,500 vehicles), consistent with commuter traffic.

■ **Northern and Southwestern outskirts**

Low traffic volumes (< 500 vehicles), proximity to slag heaps and agricultural areas, and conditions conducive to active travel.

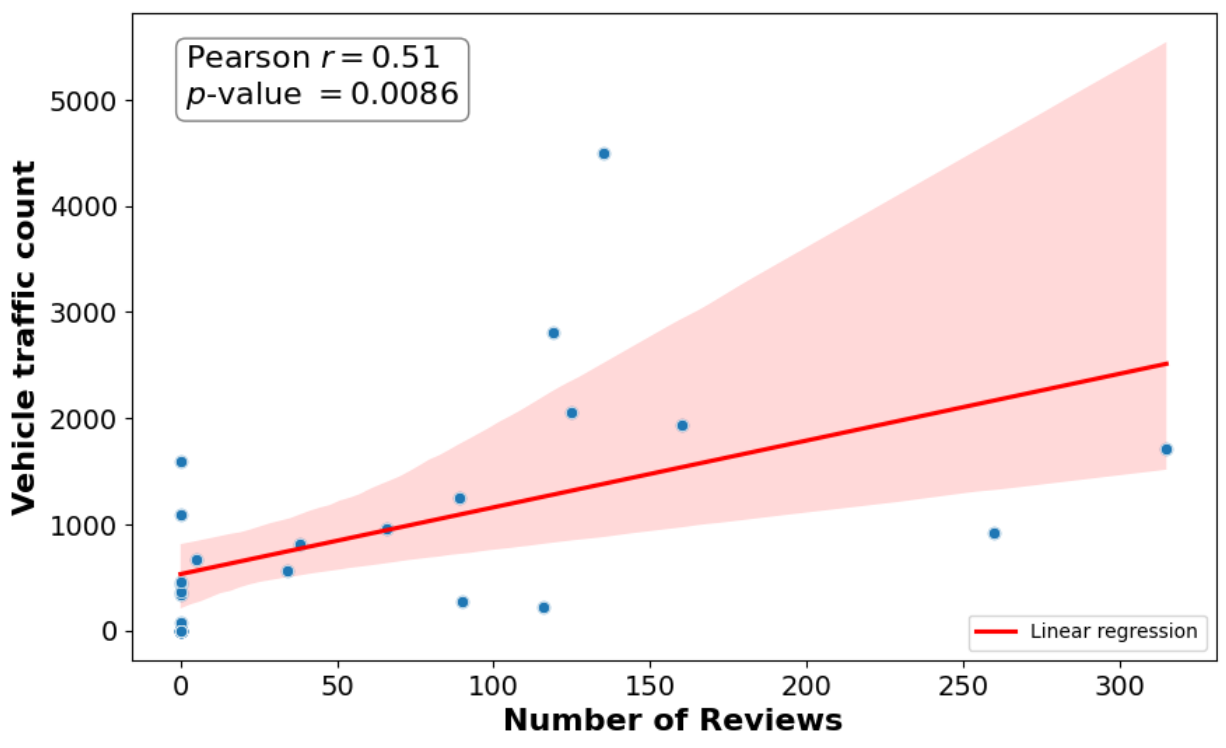
# Correlating Simulated Traffic with Online Reviews

$r = 0.51$        $p = 0.0086$

Statistically Significant Moderate Correlation ( $p < 0.05$ )

MATSim traffic volumes show a positive trend with online review counts.

→ Online review volumes show strong consistency with simulated mobility patterns.



Pearson Correlation Between Vehicle Traffic and Review Volumes

# Model Validation and Key Limitations



## Validation

- High traffic volumes and dense online review activity co-locate spatially.
- MATSim transport simulations serve as a viable proxy for modeling urban vitality.
- Offers a viable validation alternative where observed count data is lacking.



## Limitations

- Spatial bias: Online review density favors commercial zones over residential areas.
- Built environment and functional diversity influence vitality independently of traffic flow.
- Model Convergence and Scaling Limitations:  
Ongoing Sensitivity Testing

# Conclusions & Perspectives

MATSim combined with digital footprints provides a viable proxy validation method for data-scarce Peri-urban contexts.



## Future Extension

Extending the analysis to additional peri-urban contexts beyond Loos-en-Gohelle.



## Data Integration

Integrating commercial Diversity and POI data (e.g., Noufe et al., 2026).



## Calibrating Proxy Accuracy

Refine proxy scaling by linking online review activity directly to observed traffic.

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# Thanks You!

for Your Attention

# Any Questions??

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